The correlation between shutter speed and exposure is simply 1:1. Typically, exposure values (EV) are used as units to measure the exposure quantitatively. Sometimes called stops, these represent halving or doubling of the exposure. Also, multiple combinations of shutter speed and aperture can give the same exposure: halving the shutter speed doubles the exposure (1 EV more), while doubling the aperture (halving the number) increases the exposure by a factor of 4 (2 EV). For this reason, standard apertures differ by √2, or about 1.4. Thus an exposure with a shutter speed of 1/250 s and f/8 is the same as with 1/500 s and f/5.6, or 1/125 s and f/11. Keeping the shutter speed to something like in the range of 0.001 second can absolutely freeze fast moving, up close motion while 1 – 30+ second shots can be used for specialty night and low-light photos on tripods.

ISO speed

This parameter is a measure of the sensitivity your camera has to incoming light, and like shutter speed, it too correlates 1:1 with the increase or decrease in exposure. In fact, the ISO settings in cameras are a carryover from the film era of photography when ASA speeds were popular. During that time, film rolls used to be rated at 100, 400 or maybe even 1,600. This number referred to the film's light sensitivity, for higher sensitivity, you had to pick a higher value. In its present form, the ISO bit is from the standards for film sensitivity (not really an acronym as such but named so by the International Organization for Standardization), and the number refers to its rating. While film has a static rating, a single digital camera can allow you to capture images at several different ISO speeds by simply amplifying (or diminishing) the image signal in the camera. However, note that as the signal is amplified, so is the image noise, thus unlike aperture or shutter speed, higher ISO speeds dramatically increase image noise and are almost always undesirable, much like the "grainy" effects on films with increased sensitivity.

With digital cameras, common ISO speeds include 100, 200, 400 and 800, though cameras permit lower or higher values also. Effectively, what this implies is that a photo at, say, ISO 200 will take half as long to reach the same level of exposure as one taken at ISO 100 provided all remaining settings are the same. ISO speeds in the range of 50-200 typically produce acceptably low noise images in compact digital cameras, whereas with digital SLR cameras, an acceptable range can well be from 50 to 800 or even higher. Now you may wonder that since you can easily change ISO settings with every shot taken, surely you could work with settings like 297 or 839 if you wanted. Well, in theory, yes. You could pick any